



Paper ID : 261564

Printed Page: 1 of 2

Subject Code:BCAI061

Roll No:

BTECH

**(SEM. VI) THEORY EXAMINATION 2025-26**  
**CYBER FORENSIC ANALYTICS**

TIME: 3 HRS

M.MARKS: 70

**Note:** Attempt all Sections. In case of any missing data; choose suitably.

**SECTION A**

**1. Attempt all questions in brief.**

**02 x 7 = 14**

| Q no. | Question                                                               | CO | Level |
|-------|------------------------------------------------------------------------|----|-------|
| a.    | Explain the security challenges involved in eCash systems.             | 1  | K1    |
| b.    | What is cyber terrorism and cyber warfare?                             | 1  | K1    |
| c.    | Explain the importance of digital devices in cyber forensics.          | 2  | K1    |
| d.    | What are DoS and DDoS attacks?                                         | 2  | K1    |
| e.    | Explain the concepts of cyber investigation and network investigation. | 3  | K1    |
| f.    | Define evidence and digital evidence.                                  | 4  | K1    |
| g.    | Explain the concept of electronic governance (e-Governance).           | 5  | K1    |

**SECTION B**

**2. Attempt any three of the following:**

**07 x 3 = 21**

|    |                                                                                                                                                        |   |    |
|----|--------------------------------------------------------------------------------------------------------------------------------------------------------|---|----|
| a. | Explain the concept of cyber space and discuss how cyber crimes differ from traditional crimes.                                                        | 1 | K3 |
| b. | What is data carving in cyber forensics? Explain different data carving techniques used to recover deleted or hidden files.                            | 2 | K2 |
| c. | Discuss the importance of audit logs in cyber investigations. Explain the methods used for investigating audit logs to identify suspicious activities. | 3 | K3 |
| d. | Discuss the importance of contextual information in evidence analysis and cyber crime investigations.                                                  | 4 | K2 |
| e. | Explain the objectives and key features of the Information Technology Act 2000 in regulating cyber activities and cyber crimes in India.               | 5 | K2 |

**SECTION C**

**3. Attempt any one part of the following:**

**07 x 1 = 07**

|    |                                                                                                                                                                      |   |    |
|----|----------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|----|
| a. | Describe different types of cyber financial frauds, including ATM frauds, phishing, and phreaking. Suggest preventive measures against them.                         | 1 | K2 |
| b. | Explain the working of key loggers and email frauds as practical cyber crime techniques. How can individuals and organizations protect themselves from such attacks? | 1 | K3 |

**4. Attempt any one part of the following:**

**07 x 1 = 07**

|    |                                                                                                                                             |   |    |
|----|---------------------------------------------------------------------------------------------------------------------------------------------|---|----|
| a. | Explain the concept of steganography and its role in cyber crime. How do forensic investigators detect hidden information in digital media? | 2 | K2 |
| b. | Differentiate between Trojans, Backdoors and Botnets. Explain how these malware attacks affect computer systems and networks.               | 2 | K3 |



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**5. Attempt any one part of the following: 07 x 1 = 07**

|    |                                                                                                                                                  |   |    |
|----|--------------------------------------------------------------------------------------------------------------------------------------------------|---|----|
| a. | Explain the concept of profiling in cyber investigations. How are cyber criminal profiling and stylometric techniques used to identify suspects? | 3 | K2 |
| b. | Differentiate between warranted searches and warrantless searches in cyber investigations. Discuss the legal and ethical challenges involved.    | 3 | K3 |

**6. Attempt any one part of the following: 07 x 1 = 07**

|    |                                                                                                                                                    |   |    |
|----|----------------------------------------------------------------------------------------------------------------------------------------------------|---|----|
| a. | Describe the role of evidence management in cyber forensics. Explain the procedures followed during pre-search activities and on-scene activities. | 4 | K2 |
| b. | Discuss the practical applications of digital evidence analysis and network analysis in identifying and prosecuting cyber criminals.               | 4 | K2 |

**7. Attempt any one part of the following: 07 x 1 = 07**

|    |                                                                                                                                    |   |    |
|----|------------------------------------------------------------------------------------------------------------------------------------|---|----|
| a. | What are electronic signature certificates? Explain the procedure for obtaining and using them in secure online communication.     | 5 | K2 |
| b. | Describe the penalties and compensation provisions under the Information Technology Act 2000 for cyber offences and data breaches. | 5 | K3 |